KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2018
Department of Computer Science and Engineering
CSE 4207
Computer Graphics

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is computer graphics deal with? Mention some prominent applications of it. (06)
   b) Explain the geometry stage of graphics rendering pipeline with necessary illustration. (12)
   c) Find a transformation matrix that aligns line L_1, with line L_2 given by the following equations: (17)

$$L_1 = \begin{cases} \bar{v} = 2I - 3J + 4K \\ P_0 = (1,1,0) \end{cases}$$

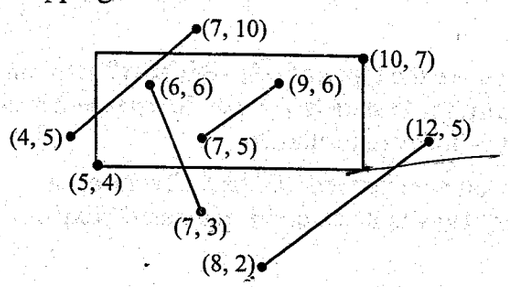
$$L_2 = \begin{cases} x = 1 + 3t \\ y = 4t \quad 0 \le t \le 1 \\ z = 2 - 5t \end{cases}$$

2. a) Derive a transformation matrix to convert an asymmetric view frustum to a Canonical View Volume (CVV). (15)
   b) What is projection? Find a parallel projection of the point (7, 8, 9) with DOP, $\bar{v} = I + 2J + K$ onto a projection plane parallel to xy plane and cuts z-axis at a distance of 1. (10)
   c) What are the parameter needed to define a View-Coordinate System (VCS)? How can a world coordinate point be converted to VCS point? (10)

3. a) Let a VCS point $p(3, 2, 6)$, projection reference point is at VCS origin, and a symmetric view volume is defined by the following parameters: (14)

$$left = right = 1,\ bottom = top = 0.8,\ near = 1,\ far = 7$$

   Find the Normalized Device Coordinate System (NDCS) value of the point p.
   b) Clip the lines presented in the following figure using Cohen-Sutherland algorithm. Also mention how CVV helps 3D clipping. (12)



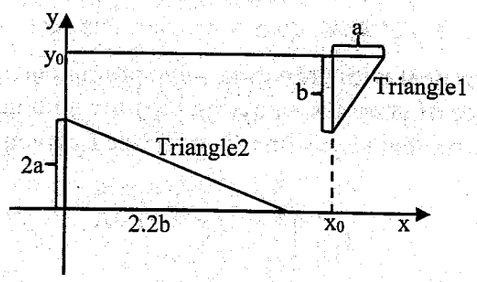
   c) Show that the path of a ray transmitted through a surface is $t = z - \left(\sqrt{1 - |z|^2}\right).n$, where (09)
   $z = \frac{\eta_1}{\eta_2}(u - (u.n))n$, u is an incoming ray, n is the surface normal, and η_1, η_2 are the refractive indices of the media.

4. a) "Phong shading is more realistic than Gouraud shading, but expensive"– Justify the statement. (12)
   b) Let a surface is illuminated by an RGB light source with light intensity (0.9, 1, 0.9), positioned at (5, 4, 2). A vertex (4, 1, 3), whose surface normal is (0, 2, –1), is viewed from (3, 2, 2). Find the intensity that the viewer will see at the vertex, if the surface diffuse and specular reflectively are (0.4, 0.3, 0.6) and (0.8, 0.8, 0.8) with shininess = 10. Consider there is no ambient light, and no attenuation of light. (15)
   c) Explain a method to find the intersection of an incoming ray with a sphere. (08)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What are the raster and vector graphics? Write the differences between vector and raster graphics. (07)
   b) Convert these homogeneous points to Cartesian: (0, 1, 3, 2) and (2, 1, 3, 4). (07)
   c) Transform Triangle1 to Triange2 in the following figure by a sequence of basic transformations. Write the transformation matrices in 2D Homogeneous Coordinates. Compute the final transformation matrix. Note that the size of the triangle has also been changed; see the following figure for the dimensions. (14)



   d) Explain the difference between "Object Coordinates", "World Coordinates", and "Camera Coordinates". (07)

6. a) What are the three major efficiency ingredients in Bresenham's line drawing algorithm? (07)
   b) Consider the line from (0, 0) to (–8, –4), use general Bresenham's line algorithm to rasterize this line. Evaluate and tabulate all the steps involved. (12)
   c) Write down the difference between DDA line drawing algorithm and Bresenham's line drawing algorithm. (06)
   d) Explain the steps in Bresenham's circle drawing algorithm. (10)

7. a) How many real values are necessary to describe the shape of a cubic rational Beizer curve in three dimensional space? Provide a short justification for you answer. (06)
   b) Prove that $R(\theta_1).R(\theta_2) = R(\theta_1 + \theta_2)$, where $R()$ is a rotation function and θ_1, θ_2 are angle of rotation. (08)
   c) Define Hermite Surface. Find the geometry vector G_{Hx} for Hermite Surface. Also find the first of the vecrtor. (10)
   d) Construct the Beizer,curve of order 3 with 4 polygon vertices A(0, 1), B(3, 3), C (4, 5) and D (7, 5) (11)

8. a) What is fractals? What are the types of self-similarity? Explain with example(s). (08)
   b) Explain z-Buffer algorithm in details for visible surface detection. What are the advantages and disadvantages of z-Buffer algorithm? (13)
   c) What is continuity? Compare between G^1 and C^1 continuity. (06)
   d) What is Koch Curves? How is Koch curve generated? Explain with suitable example(s)? (08)